

ESPOO TAPIOLA, Finland

WMO: 029851

Lat: 60.1781N

Lon: 24.7872E

Elev: 25

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.0	2.4	-7.7	3.6	-2.6	-1.7	5.0	4.1	22.7	33.0	20.7	32.5	6.2	10	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	80.3	66.9	77.0	65.1	74.0	63.1	69.3	76.8	67.1	74.4	64.9	71.7	8.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.5	97.5	73.5	64.3	90.1	70.6	62.0	83.2	68.4	33.4	76.9	31.5	74.5	29.9	71.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.3	15.5	-5.9	83.8	8.2	4.5	-11.8	87.1	-16.6	89.7	-21.2	92.3	-27.1	95.5		
(5)			-6.3	70.5	8.1	2.5	-12.1	72.4	-16.9	73.8	-21.5	75.3	-27.4	77.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	22.9	25.4	30.6	40.3	51.6	58.5	64.6	61.6	53.7	42.5	36.3	29.8		
	DBStd	15.71	10.19	10.01	7.08	5.75	7.12	5.63	5.53	4.83	5.47	6.07	7.31	8.98		
	HDD50	3803	841	690	603	295	63	5	0	0	28	240	411	627		
	HDD65	8042	1306	1110	1068	740	417	208	77	126	340	697	861	1092		
	CDD50	1338	0	0	0	5	114	260	452	361	138	8	0	0		
	CDD65	100	0	0	0	0	2	12	64	22	0	0	0	0		
	CDH74	668	0	0	0	0	25	87	435	121	0	0	0	0		
	CDH80	100	0	0	0	0	1	7	78	14	0	0	0	0		
(14) Wind	WSAvg	7.9	8.0	7.9	8.4	7.4	7.7	8.0	7.1	7.1	7.3	8.1	8.6	9.3		
(15) Precipitation	PrecAvg	23.7	1.7	1.3	1.2	1.3	1.4	2.0	2.2	2.9	2.4	2.7	2.6	2.0		
	PrecMax	33.0	3.3	3.9	2.2	2.8	2.9	4.9	6.0	6.5	6.1	7.0	6.7	5.0		
	PrecMin	14.2	0.0	0.0	0.2	0.3	0.2	0.4	0.1	0.0	0.7	0.6	0.2	0.2		
	PrecStd	4.0	0.9	0.9	0.6	0.7	0.8	1.1	1.6	1.6	1.3	1.4	1.2	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.2	43.3	49.6	64.8	77.3	80.3	86.1	82.3	70.0	58.0	51.2	47.5		
		MCWB	38.7	38.2	41.2	50.8	61.4	63.3	68.4	68.4	61.4	53.0	49.0	45.8		
	2%	DB	38.6	39.3	45.4	60.6	72.7	76.4	82.1	77.4	66.8	55.3	48.9	44.6		
		MCWB	37.6	37.1	38.9	47.8	58.0	62.8	67.8	66.0	60.1	52.3	47.6	42.6		
	5%	DB	37.0	37.4	42.1	55.4	69.2	73.2	78.8	73.7	64.6	53.6	46.8	42.2		
		MCWB	36.4	36.2	37.3	46.2	56.4	61.1	66.8	63.7	58.9	51.5	45.6	40.7		
	10%	DB	35.1	35.8	38.8	51.2	65.4	69.7	75.7	70.6	62.3	51.7	45.4	40.0		
		MCWB	34.5	34.9	35.9	42.8	54.5	59.0	65.2	61.9	58.0	49.6	44.2	38.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.1	38.7	43.1	51.7	64.3	68.2	71.6	71.4	63.2	55.0	49.7	46.4		
		MCDB	39.8	40.9	48.0	63.2	74.5	75.6	80.2	78.7	66.7	56.6	50.7	47.2		
	2%	WB	37.8	37.6	39.9	49.0	60.5	64.8	70.1	67.7	61.7	53.5	47.9	43.1		
		MCDB	38.6	39.0	43.6	58.0	68.4	73.8	78.0	73.6	65.1	54.6	48.8	44.2		
	5%	WB	36.4	36.3	38.1	46.4	58.2	62.3	68.6	65.6	60.3	51.8	45.9	40.8		
		MCDB	37.0	37.3	40.9	54.2	66.3	70.0	76.5	71.5	63.3	53.0	46.7	41.9		
	10%	WB	34.5	35.0	36.4	44.0	55.5	60.3	66.4	63.6	58.7	50.0	44.4	38.8		
		MCDB	35.0	35.8	38.6	49.7	64.5	67.9	74.1	68.5	61.4	51.3	45.3	39.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	8.5	11.5	14.9	17.0	15.2	15.3	14.7	12.7	10.6	7.0	7.4		
		MCDBR	6.1	8.1	14.7	22.5	22.7	20.2	19.0	18.4	15.5	12.6	7.6	8.3		
	5% WB	MCWBR	5.7	6.7	9.5	12.5	11.2	10.0	8.0	8.6	9.4	9.6	7.2	8.1		
		MCWBR	6.1	7.3	12.5	20.7	19.3	17.9	17.2	15.8	14.0	10.1	7.3	7.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.279	0.313	0.335	0.360	0.358	0.356	0.378	0.372	0.344	0.334	0.307	0.264		
		taud	2.139	2.191	2.249	2.344	2.417	2.477	2.438	2.454	2.519	2.394	2.216	2.088		
	Ebn at Noon	Ebn at Noon	150	212	249	266	276	278	268	260	249	207	138	106		
		Edh at Noon	14	23	30	33	33	31	32	29	23	19	13	10		
(44) All-Sky Solar Radiation	RadAvg	78	259	732	1220	1667	1732	1671	1295	789	344	96	47			
	RadStd	14	50	94	128	180	135	152	133	102	54	17	7			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		+0.77	N/A	N/A	N/A	N/A	N/A	-248	N/A	N/A	N/A

Nomenclature: See separate page